

Defects and non-stoichiometry

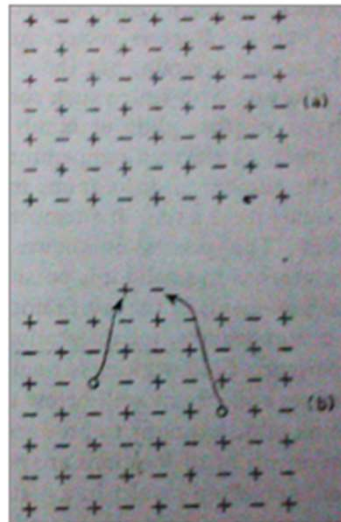
Perfect crystal can be obtained, hypothetically, only at absolute zero; at all real temperatures, crystals are imperfect.

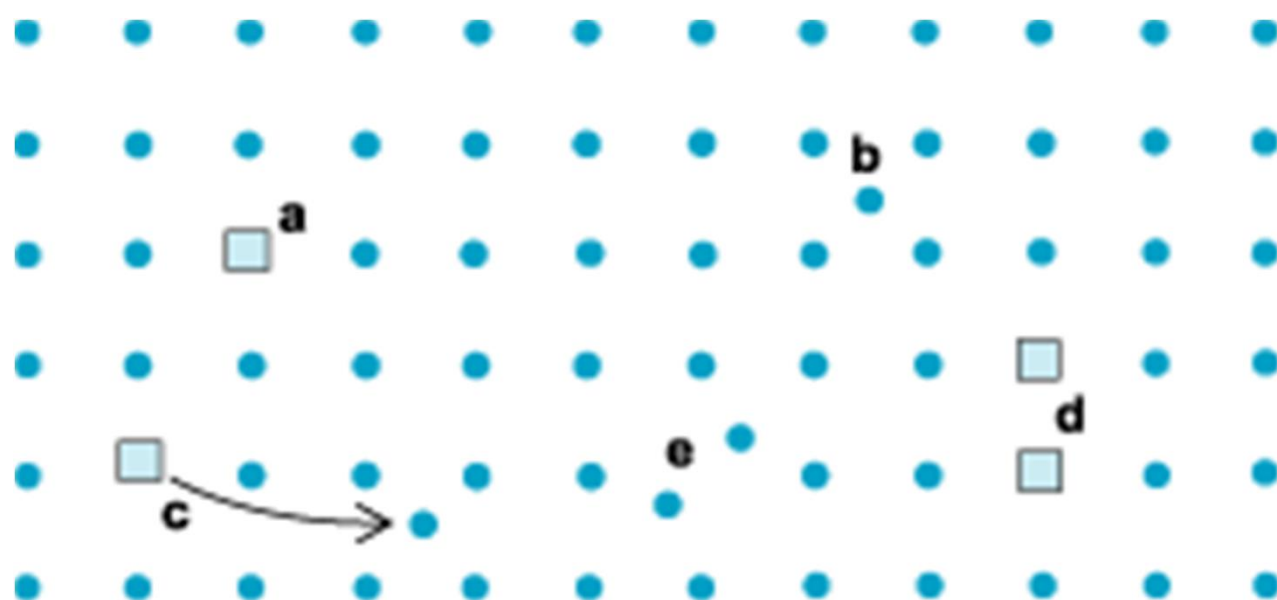
Perfect crystals do not exist; even the best crystals have 1ppb defects.

Defects are imperfections in the regular repeating pattern.

Defect

The term defect is used to describe both missing and misplaced atoms (or ions) from positions in the ideal crystal lattice (also combinations of these) and to systems involving incorporation of foreign ions within the solid.





Key:

a = vacancy (Schottky defect)

b = interstitial

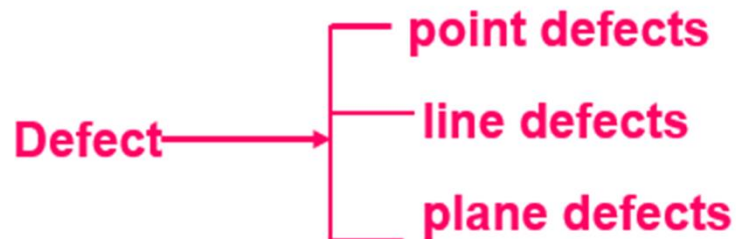
c = vacancy-interstitial pair (Frenkel defect)

d = divacancy

e = split interstitial

 **= vacant site**

Types of Defects



Types of Defects

Stoichiometric defect

Stoichiometric defects in which the crystal composition is unchanged on introducing the defects.

Non-stoichiometric defect

Non-stoichiometric defects which are a consequence of a change in crystal composition.

Schottky and Frenkel Defects

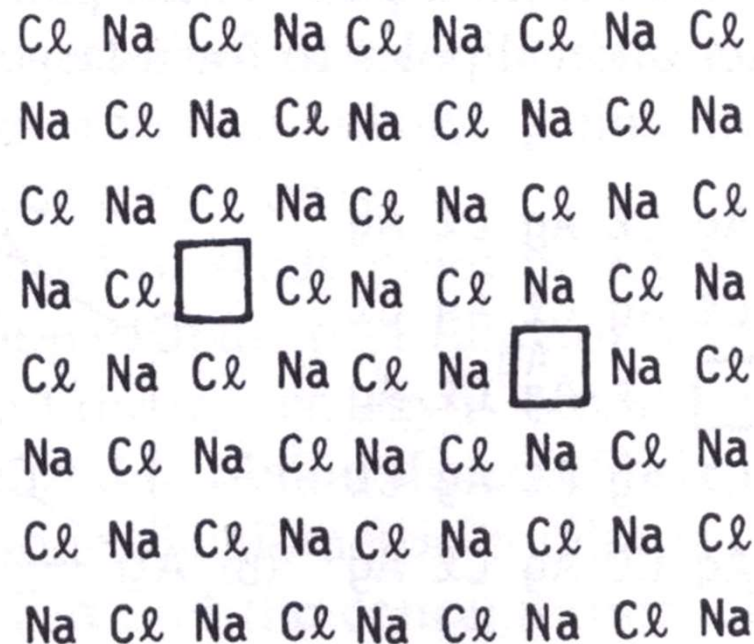
Schottky defects and Frenkel defects named after the research workers who first suggested their existence. They are stoichiometric defects.

Schottky defects

The Schottky defect is a stoichiometric defect in ionic crystals. An ionic crystal containing Schottky defects has an equal number of cation and anion vacancies. Such defects can be regarded as displacement of cations and anions from a number of random lattice positions, to the crystal surface as shown in Fig. This is illustrated for an ionic solid.

Atoms from a metal crystal, or molecules from within a molecular crystal, may be similarly transferred to the surface leaving lattice vacancies.

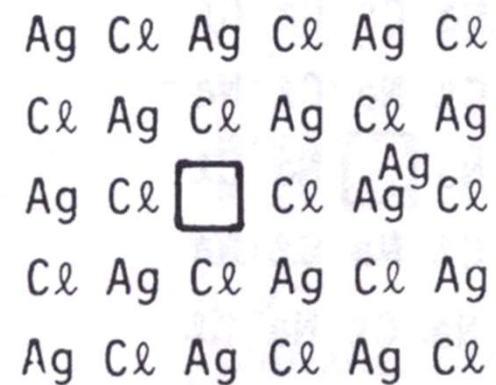
The Schottky defect is the principal point defect in the alkali halides and is shown for NaCl in Fig.



There are equal numbers of anion and cation vacancies in order to preserve local electro neutrality as much as possible, both inside the crystal and at the surface of the crystal.

Frenkel Defect

Frenkel defect is also a stoichiometric defect and involves an atom/ion displaced off its lattice site into an interstitial site that is normally empty. A Frenkel defect is formed (Fig.1) by the transference of an ion/atom from a normal lattice position to an interstitial (i.e. `between lattice points`) position in a different region of the crystal.



Frenkel Defect

It is usually the cation which is transferred in this way since this is often the smaller ion and thus more easily accommodated at the inter-lattice point..

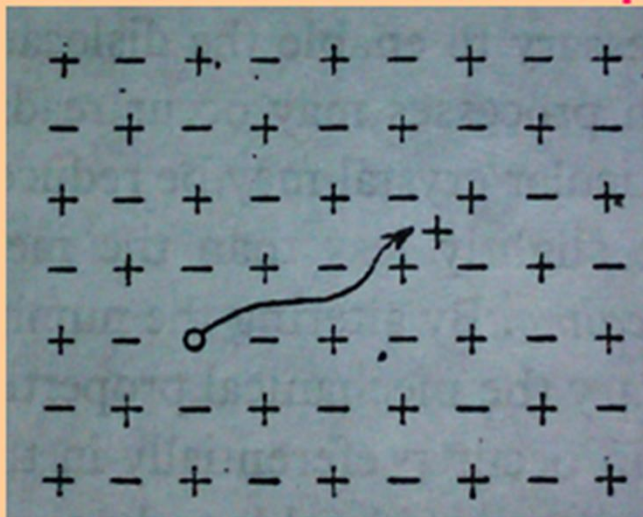


Fig 1. Frenkel defect formation:
An ion, usually a cation, removed from the lattice point and transferred to an interstitial lattice position.

There is probably some covalent interaction between the interstitial Ag^+ ion and its four Cl^- neighbours which acts to stabilize the defect and give Frenkel defects, in presence of Schottky defects, in AgCl .

On the otherhand, Na^+ ion, with its 'harder', more cationic character, would not find much comfort in a site which was tetrahedrally surrounded by four other Na^+ ions, Frenkel defects therefore do not occur to any significant extent in NaCl .

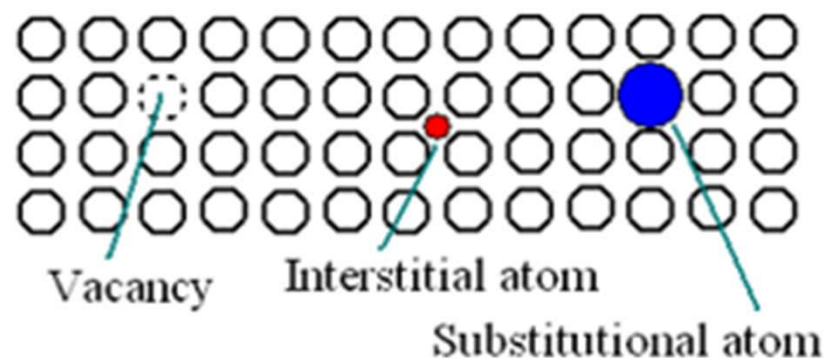
Calcium fluoride, CaF_2 , also has predominantly Frenkel defects but in this case it is the anion, F^- , which occupies the interstitial site.

Other materials with fluorite and antifluorite structures have similar defects, e.g. ZrO_2 (O^{2-} interstitial) and Na_2O (Na^+ interstitial).

Types of Defects

The size and shape of the defect can be used as a basis for classification:

Point defects: point defects involve only one atom or site, e.g. vacancies or interstitials, although the atoms immediately surrounding the defect are also somewhat perturbed.



Schematic depiction of various point defects

Types of Defects

Line defect: *Line defects, i.e. dislocations, are effectively point defects in two dimensions but in the third dimension the defect is very extensive or infinite.*

Plane defects: In plane defects, whole layers in a crystal structure can be defective.

Sometimes the name extended defects is used to include all those who do not come in the category of point defects.

LINE IMPERFECTIONS

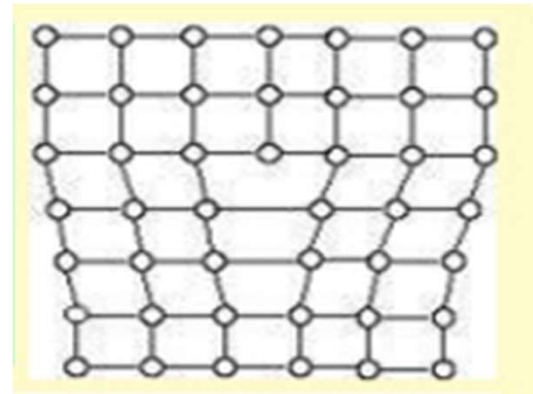
- ✚ The defects, which take place due to dislocation or distortion of atoms along a line, in some direction are called as '*line defects*'.
- ✚ Line defects are also called dislocations. In the geometric sense, they may be called as 'one dimensional defects'.
- ✚ A dislocation may be defined as a disturbed region between two substantially perfect parts of a crystal.
- ✚ It is responsible for the phenomenon of slip by which most metals deform plastically.

The two types of dislocations are,

- ✚ Edge dislocation
- ✚ Screw dislocation

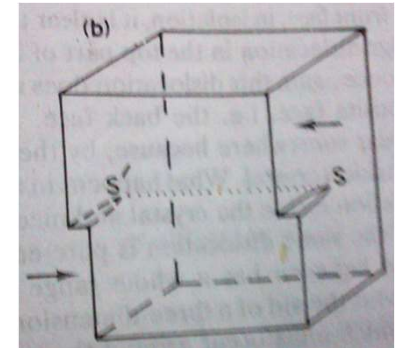
EDGE DISLOCATION

- An edge dislocation is a defect where an extra half-plane of atoms is introduced through the crystal, distorting nearby planes of atoms.



Screw Dislocations

If the misalignment of ion or presence of vacancies along a line shifts a block of ions gradually downward or upwards, a screw like deformation, which is called screw dislocation.

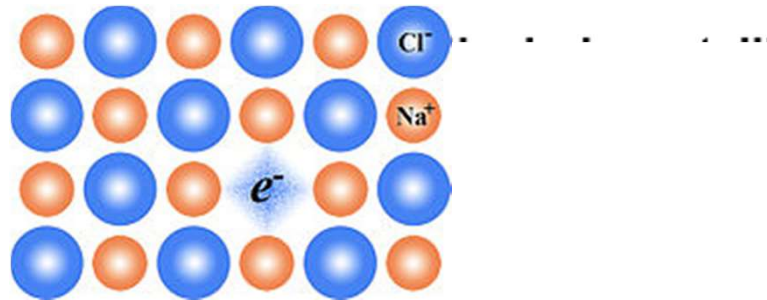


Color centers

Colour centres are an example of extrinsic defects, i.e. those caused by the presence of "impurity", in this case just an excess of one atom type.

Color centers can occur naturally in compounds (particularly metallic oxides) because when heated to high temperature the ions become excited and are displaced from their normal crystallographic positions, leaving behind some electrons in the vacated spaces.

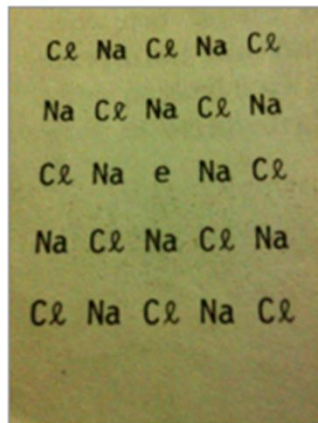
An F-center, *Farbe* center or color center (from the original German *Farbzentrum*; *Farbe* means *color*, and *zentrum* center) is a type of crystallographic defect in which an anionic vacancy in a crystal is filled by one or more electrons. Electrons in such a vacancy tend to absorb light in the visible spectrum such that a material that is usually transparent becomes colored. This is used to identify many compounds, especially zinc oxide (yellow).



F-center in an NaCl crystal

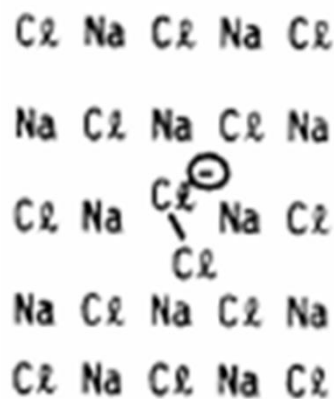
The greater the number of F-centers, the more intense is the color of the compound.

An H-center (a halogen interstitial) is in a sense the opposite to an F-center, and hence the two can combine and clear the lattice of a defect. This process can be photoinduced, e.g., using a laser.



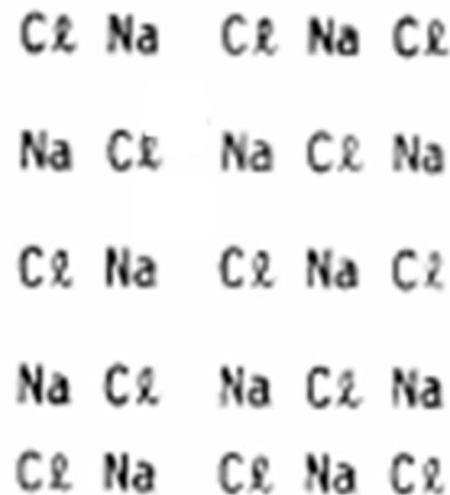
F-centre

+



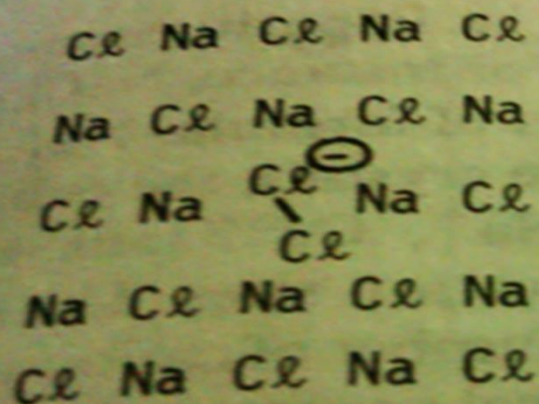
H-centre

=

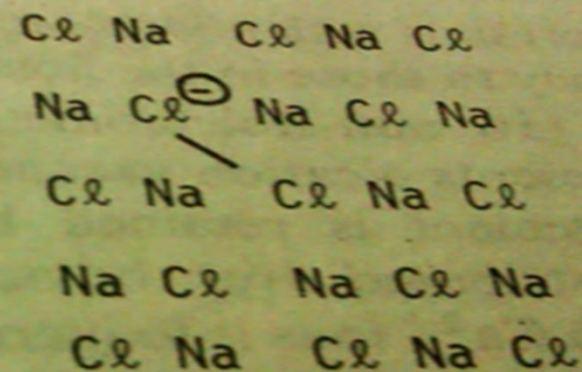


Perfect crystal

Many other colour centres have been characterized in alkali halide crystals; two of these, the H-centre and V-centre, are shown in Fig. 2. Both contain the chloride molecule ion, Cl_2^- , but this occupies one site in the H-centre and two sites in the V-centre; in both cases the axis of the Cl_2^- ion is parallel to the direction.



(a)



(b)

Fig. 9.6 (a) H-centre and (b) V-centre in NaCl

A perfect crystal is the one that contains no point, linear, or planar imperfections.

A crystal in which the regular, periodic structure is interrupted by various defects is imperfect crystal.

Why perfect crystal does not exist?

If a crystal is of infinite size, then there are an infinite number of atomic sites which could have a defect, and (because absolute zero is an infinitely improbable state) there will be at least one which does. Therefore there are no perfect crystals.